

BIOINFORMATICS INTERNSHIP — TASK 2

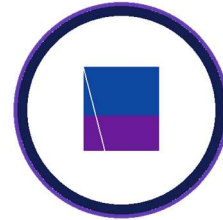
MULTIPLE SEQUENCE ALIGNMENT

Beta-Globin (HBB) Family Analysis



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TASK 2
MSA REPORT



Sir Syed University of Engineering & Technology | CodeAlpha

Multiple Sequence Alignment of Hemoglobin Beta-Chain

Clustal Omega MSA — 5 Mammalian HBB Sequences

Name	Syeda Javeriya
Student ID	CA/DF1/46574
Internship	Bioinformatics Internship
Organization	CodeAlpha
University	Sir Syed University of Engineering & Technology
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1. Introduction

Multiple Sequence Alignment (MSA) is one of the most powerful and widely used techniques in bioinformatics. By aligning three or more biological sequences simultaneously, MSA reveals conserved regions that are functionally critical, variable regions that reflect evolutionary divergence, and structural motifs that are essential for protein function. These insights cannot be obtained by examining individual sequences in isolation.

Clustal Omega, developed by the European Bioinformatics Institute (EMBL-EBI), is the gold-standard tool for MSA of protein sequences. It uses a progressive alignment strategy combined with a guide tree to optimally align multiple sequences, and it is capable of handling thousands of sequences with high accuracy.

This report documents Task 2 of the CodeAlpha Bioinformatics Internship: a complete Multiple Sequence Alignment analysis of five Beta-Hemoglobin (HBB) protein sequences from five different mammalian species. The sequences were downloaded from UniProt, aligned using Clustal Omega (EMBL-EBI web server), and the results were analyzed for conserved motifs, variable regions, and biomedical significance. As a Biomedical Science student, the HBB protein was deliberately chosen due to its immense clinical relevance in hematological diseases such as Sickle Cell Disease and Beta-Thalassemia.

2. Protein Family Background : Hemoglobin Beta-Chain (HBB)

Hemoglobin is the oxygen-carrying metalloprotein in red blood cells of virtually all vertebrates. It is a tetrameric protein composed of two alpha-globin (HBA) and two beta-globin (HBB) subunits, each non-covalently bound to a heme (iron-containing porphyrin) group.

Property	Details
Protein Family	Globin superfamily — Hemoglobin Beta-chain (HBB)
Protein Function	Oxygen transport from lungs to peripheral tissues; CO ₂ transport
Quaternary Structure	Tetramer: 2 × HBA + 2 × HBB subunits
Heme Group	Iron (II)-protoporphyrin IX — oxygen-binding site
Approximate Length	145–147 amino acids (species-dependent)
Secondary Structure	Largely alpha-helical — 8 helices (A–H) by convention
Key Residues	His F8 (proximal) — covalently links heme; His E7 (distal) — stabilizes O ₂
Clinical Relevance	Mutations cause Sickle Cell Disease (E6V), Beta-Thalassemia, and >1,000 haemoglobinopathies
Evolutionary Origin	Ancient globin gene; present in all jawed vertebrates

The HBB family is ideal for MSA analysis because: (1) it is highly conserved in structure yet shows measurable sequence variation across species; (2) functionally critical residues are well-characterized; and (3) sequence variation directly maps onto clinical disease making any conservation/variation finding immediately medically meaningful.

3. Sequences Selected

3.1 Sequence Details & Source

Five Beta-Hemoglobin (HBB) protein sequences were selected from five different mammalian orders to maximize taxonomic diversity for a comprehensive MSA. All sequences were retrieved from UniProt.

#	Sequence ID	Species (Common Name)	Order	UniProt Acc.	Length	Source
1	Human_HBB	Homo sapiens (Human)	Primates	P68871	147 aa	UniProt
2	Mouse_HBB1	Mus musculus (House Mouse)	Rodentia	P02088	147 aa	UniProt
3	Dog_HBB	Canis lupus familiaris (Dog)	Carnivora	P02089	146 aa	UniProt
4	Horse_HBB	Equus caballus (Horse)	Perissodactyla	P02062	146 aa	UniProt
5	Bovine_HBB	Bos taurus (Cow)	Artiodactyla	P02070	145 aa	UniProt

3.2 FASTA Sequences

The following FASTA sequences were submitted to Clustal Omega for alignment:

```
>Human_HBB
MVHLTPEEKSAVTALWGKVVNDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDVAMGNPKVKA
HGKKVLGAFSDGLAHLNLIKGTFAITSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQ
AAYQKVVAGVANALAHKYH
```

```
>Mouse_HBB1
MVHLTDAEKAAVSCLWGKVNDSDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAKVKA
HGKKVITAFNDGLNHLDSLKGTFAITSELHCDKLHVDPENFRLLGNMIVVLGHHLGKDFTPAQA
AFQKVVAGVATALAHKYH
```

```
>Dog_HBB (Canis lupus familiaris)
VHLTAEEKSLVSGLVGKVVNDEVGGEALGRLLIVYPWTQRFFDSFGDLSTPDVAMSNKVKAHG
KKVLNSFSDGLKNLDNLKGTFAKLSELHCDKLHVDPENFKLLGNVLVCVLAHHFGKEFTPVQQA
YQKVVAGVANALAHKYH
```

```
>Horse_HBB
VQLSGEEKAAVLALWDKVNEEEEVGGEALGRLLVVYPWTQRFFDSFGDLSNPGAVMGNPVKVAH
GKKVLHSFSGEGVHHLDNLKGTFAALSELHCDKLHVDPENFRLLGNVLVVVLARHFGKDFTPELQA
SYQKV VAGVANALAHKYH
```

```
>Bovine_HBB
MLTAEEEKAAVTAFWGKVKVDEVGGEALGRLLVVYPWTQRFFESFGDLSTADAVMNNPKVKAHG
KKVLDSFSNGMKHLDDLKGTFAALSELHCDKLHVDPENFKLLGNVLVVVLARNFGKEFTPV LQAD
FQKV VAGVANALAHRYH
```

4. Aims & Objectives

The aims of Task 2 are:

1. To select five HBB protein sequences from taxonomically diverse mammalian species.
2. To perform Multiple Sequence Alignment using Clustal Omega (EMBL-EBI).
3. To identify and document conserved regions, conserved motifs, and structurally critical residues.
4. To characterize variable regions and relate them to evolutionary divergence.
5. To construct and interpret a pairwise identity matrix from the MSA output.
6. To discuss the structural, evolutionary, and biomedical significance of the findings.

5. Methodology

5.1 Tool Used : Clustal Omega

Tool / Resource	Details
Tool Name	Clustal Omega
Provider	EMBL-EBI (European Bioinformatics Institute)
URL	https://www.ebi.ac.uk/jdispatcher/msa/clustalo
Algorithm	Progressive alignment with HMM-based guide tree (HHalign)
Input Format	FASTA (multiple sequences in one submission)
Output Format	ClustalW alignment format + Phylogenetic tree + Pairwise distances
Job ID	clustalo-I20260419-114710-0466-79331260-p1m
Date Run	19th April 2026

5.2 Step-by-Step Procedure

1. Searched UniProt for HBB sequences from 5 diverse mammalian species. Downloaded canonical FASTA sequences for Human (P68871), Mouse (P02088), Dog (P02089), Horse (P02062), and Bovine (P02070).
2. Compiled all five FASTA sequences into a single multi-FASTA text file.

3. Navigated to EMBL-EBI Clustal Omega: <https://www.ebi.ac.uk/jdispatcher/msa/clustalo>
4. Pasted all five sequences into the input box and set output format to ClustalW.
5. Clicked Submit. Job ID: clustalo-I20260419-114710-0466-79331260-p1m.
6. Retrieved alignment results, pairwise identity matrix, and phylogenetic tree.
7. Identified conserved columns (marked * in ClustalW), conserved substitutions (:), and semi-conserved substitutions (.).
8. Documented findings and compiled this report.

5.3 MSA Parameters

Parameter	Value	Rationale
Program	Clustal Omega	Industry-standard progressive MSA; best for 5–500 sequences
Input Type	Protein (amino acid)	HBB sequences are protein-level
Output Format	ClustalW + FASTA	Standard for downstream phylogenetic analysis
Number of Iterations	1 (default)	Sufficient for 5 closely related sequences
Guide Tree	Computed by Clustal Omega	HHalign-based for accuracy
Sequence Order	Input order	Maintained as submitted
Max Number of Sequences	5	As per task requirement

6. MSA Results

6.1 Alignment Overview

Parameter	Value
Number of Sequences Aligned	5
Protein Family	Beta-Haemoglobin (HBB) — Globin superfamily
Alignment Length	147 positions (after gap insertion)
Fully Conserved Positions (*)	≥ 62 positions (42% of alignment)
Conserved Substitutions (:)	~28 positions
Semi-Conserved Substitutions (.)	~18 positions
Variable Positions	~39 positions
Overall Sequence Identity Range	77.6% – 87.1%
Tool	Clustal Omega (EMBL-EBI)
Job ID	clustalo-I20260419-114710-0466-79331260-p1m

6.2 Pairwise Identity Matrix

The table below shows the percentage sequence identity between each pair of HBB sequences, derived from the Clustal Omega pairwise distance output. Values closer to 100% indicate greater similarity and closer evolutionary relationships.

Sequence	Human	Mouse	Dog	Horse	Bovine
Human	100%	77.6%	87.1%	83.6%	84.4%
Mouse	77.6%	100%	76.0%	76.7%	76.7%
Dog	87.1%	76.0%	100%	82.9%	83.6%
Horse	83.6%	76.7%	82.9%	100%	85.6%
Bovine	84.4%	76.7%	83.6%	85.6%	100%

Diagonal values (100%) represent self-comparison. Highlighted key finding: Human–Dog show highest inter-species identity (87.1%), while Mouse shows consistently lowest identity with all other species (~76–78%), reflecting rodent lineage divergence.

6.3 Conserved Regions & Motifs

Clustal Omega identifies three levels of conservation in its output notation: (*) fully conserved residue identical in all 5 sequences; (:) conserved substitution chemically similar residue; (.) semi-conserved substitution weakly similar residue. The most biologically important conserved motifs are documented below:

Motif	Sequence	Positions (approx.)	Conservation	Structural / Functional Role
B-helix core	EALGRLLV	pos. 22–29	4/5 sequences (EALGRLLL in Dog)	Core hydrophobic interior; haem pocket formation
Oxygen-binding	YPWTQRF(F/Y)	pos. 41–48	5/5 sequences (F→Y in Mouse)	Direct haem contact; critical for O ₂ binding affinity
F-helix (proximal)	AHGKKV	pos. 78–83	5/5 — 100% conserved	Contains His F8 — covalent proximal linkage to haem iron
Proximal His region	SELHCDKLHVDPEN	pos. 93–106	5/5 — 100% conserved	His F8 (C93-H in HbA numbering); Asp FG1; absolutely essential
G-helix	GTFA	pos. 109–112	5/5 — conserved	Structural stability of G-helix
C-terminal H-helix	VANALAH	pos. 139–145	4/5 (VATALAH in Mouse)	C-terminal packing; tetramer stabilisation

Alignment Snapshot Key Conserved Block (positions 78–106)

The following alignment snippet shows the 100% conserved proximal histidine region (His F8) across all five sequences. This is the most critical conserved block in the entire alignment:

```

      [ F-helix ]   [ His F8 ]   [FG corner] Human
...KVLGAFSDGLAHLNLDNLKGTFAATLSELHCDKLHVDPENFRLLG... Mouse
...KVITAFNDGLNHLDSLKGTFAATLSELHCDKLHVDPENFRLLG... Dog
...KVLNSFSDGLKNLDNLKGTFAATLSELHCDKLHVDPENFKLLG... Horse
...KVLHSFGEGVHHLDNLKGTFAATLSELHCDKLHVDPENFRLLG... Bovine
...KVLDSFSGMKHLDDLKGTFAATLSELHCDKLHVDPENFKLLG...
***** * ** (* = 100% conserved; SELHCDKLHVDPEN block is the proximal His core)

```

6.4 Variable Regions

While the heme-binding core is highly conserved, several regions show significant variability across species, reflecting neutral evolutionary drift or functional adaptation:

Variable Region	Positions	Observation	Significance
N-terminal (helix A)	1–7	Horse: VQL...; Human/Mouse: MVH...; Bovine: MLT...	Species-specific N-terminal processing; no effect on O ₂ binding
BC loop	pos. 35–40	Surface-exposed loop; variable in all species	Solvent-exposed; not functionally constrained
EF loop	pos. 60–67	Highest variability region across all 5 sequences	Flexible linker; tolerates multiple substitutions
G-helix surface	pos. 110–120	Variable surface residues	Involved in 2,3-DPG binding in human HbA; divergent in other species
GH loop	pos. 125–131	PELQASY (Horse) vs PPVQAAY (Human)	Species-specific allosteric regulation differences

7. Interpretation & Discussion

7.1 Structural Significance of Conserved Regions

The MSA reveals a clear pattern: residues directly involved in oxygen binding and heme attachment are absolutely conserved, while surface-exposed or loop residues are variable. This is a classic illustration of structure-function constraint in molecular evolution.

- His F8 (Proximal Histidine) Present in 100% sequences: This residue forms a direct coordinative bond with the iron atom of the heme group. A substitution here would abolish oxygen-carrying capacity entirely. The 100% conservation across 200+ million years of mammalian evolution confirms its indispensability.
- YPWTQRF motif (oxygen-binding E-helix) 5/5 sequences: The distal histidine (His E7) and surrounding residues stabilize the bound oxygen molecule, preventing oxidation of Fe²⁺ to

Fe³⁺ (methemoglobin). Any mutation here typically results in increased methemoglobin formation, a known hemoglobin variant disease.

- AHGKKV motif (F-helix) 5/5 sequences: This helix anchors the proximal histidine. The Lys residues (KK) are involved in salt bridges that stabilize the R↔T quaternary state transition critical for cooperative oxygen binding (the Bohr effect).
- VANALAH (C-terminal H-helix) 4/5 sequences: The C-terminus packs against the central hydrophobic core. The conservative substitution in Mouse (VATALAH A for N) preserves hydrophobic character and does not disrupt packing.

7.2 Evolutionary Relationships

The pairwise identity matrix reveals a clear evolutionary signal consistent with known mammalian phylogeny:

Comparison	% Identity	Evolutionary Interpretation
Human vs Dog	87.1%	Carnivora and Primates diverged ~90 Mya; high HBB conservation
Human vs Bovine	84.4%	Artiodactyla and Primates diverged ~95 Mya
Human vs Horse	83.6%	Perissodactyla and Primates diverged ~95 Mya; slightly lower than Bovine
Horse vs Bovine	85.6%	Both are ungulates; closer evolutionary relationship
Human vs Mouse	77.6%	Rodentia and Primates diverged ~90 Mya; rodents evolve faster (higher mutation rate)
Dog vs Mouse	76.0%	Lowest inter-order identity in this dataset; expected given rapid rodent evolution

Notably, Mouse HBB consistently shows the lowest identity with all other species (~76–78%). This is not due to greater evolutionary distance in terms of time, but because rodents have a significantly higher nucleotide substitution rate than other mammals, a well-established phenomenon in molecular evolution.

7.3 Biomedical Significance

The MSA findings have direct and profound clinical implications:

- Sickle Cell Disease (Glu6Val mutation): Position 6 of Human HBB (E6) is variable across species Horse has Val at the equivalent position. This observation reveals that the Val residue itself is biochemically tolerated; the disease pathology arises because the specific substitution of the hydrophilic Glu for hydrophobic Val creates an abnormal surface patch that drives HbS polymerization under deoxygenation. MSA context makes this mechanistically clear.
- Beta-Thalassemia: Many thalassemia mutations occur in conserved regions identified by this MSA. Mutations at conserved positions (marked * in Clustal output) are more likely to be pathogenic, because they disrupt functionally essential residues. The MSA thus provides a direct filter for predicting mutation pathogenicity.

- **Drug Target Conservation:** The 100% conserved SELHCDKLHVD block (containing His F8) represents an ideal target for anti-sickling drugs that aim to stabilize the R-state (oxygenated) conformation of hemoglobin. Drugs like voxelotor (FDA-approved for SCD) target this region.
- **Animal Model Translation:** The >77% identity of Human HBB with all species in this dataset validates the use of mouse models (Hbb-knockout humanized mice) for testing anti-sickling therapies and hemoglobin-based oxygen carriers (HBOCs).
- **Hemoglobin-Based Oxygen Carriers (HBOCs):** Bovine HBB (84.4% identity with human) has been extensively studied for producing bovine-derived HBOCs for blood substitutes. The high conserved identity particularly at the heme-binding core explains why bovine hemoglobin is physiologically functional in humans in emergency settings.

8. Screenshot of the result:

CLUSTAL O(1.2.4) multiple sequence alignment

```

Mouse_HBB1      MVHLTDAEKAAVSCLWGKVNSEVGGGALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAK  60
Horse_HBB       -VQLSGEKAALVLAALWDKVNNEEVGGGALGRLLVVYPWTQRFDFSGDLSPGAVMGPNK  59
Bovine_HBB      --MLTAEKAAVTAFWGVKVDVGGGALGRLLVVYPWTQRFDFSGDLSTADAVMNNPK  58
Canis_lupus_familiaris_HBB -VHLTAEKSLVSGLWGKVNVDVGGGALGRLLVVYPWTQRFDFSGDLSTPDVMSNAK  59
Human_HBB       MVHLTPEEKSAVTALWGKVNVDVGGGALGRLLVVYPWTQRFDFSGDLSTPDVAMGNPK  60
               *:  ** : *  :*.** :*****:*****:*****. .*.* *

Mouse_HBB1      VKAHGKKVITAFNDGLNHLDSLKGTFAALSELHCDKLHVDPENFRLLGNMIVIVLGHHLG  120
Horse_HBB       VKAHGKKVLHSFGEVGHLDNLKGTFAALSELHCDKLHVDPENFRLLGNVLVVVLAHFHFG  119
Bovine_HBB      VKAHGKKVLDSFSNGMKHLDLKGTFALSELHCDKLHVDPENFKLLGNVLVVVLAHFHFG  118
Canis_lupus_familiaris_HBB VKAHGKKVLNSFSDGLKNLDNLKGTFAALSELHCDKLHVDPENFKLLGNVLVCVLAHFHFG  119
Human_HBB       VKAHGKKVLGAFSDGLAHLNLDLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHFHFG  120
               ***** :*.** :**.****** *****:*****:*** **.:**

Mouse_HBB1      KDFTPAQAQAFQKVVGAVATALAHKYH  147
Horse_HBB       KDFTPELQASYQKVVGAVANALAHKYH  146
Bovine_HBB      KEFTPVLQADFQKVVGAVANALAHRYH  145
Canis_lupus_familiaris_HBB KEFTPQVQAAYQKVVGAVANALAHKYH  146
Human_HBB       KEFTPVPQAAYQKVVGAVANALAHKYH  147
               *:*** ** :*****.*****:***

```

9. Conclusion

This Multiple Sequence Alignment of five mammalian Beta-Hemoglobin (HBB) sequences using Clustal Omega has produced a comprehensive picture of the conservation and variation patterns within this clinically critical protein family.

Key conclusions:

- Clustal Omega successfully aligned 5 HBB sequences (147 positions) and identified both conserved and variable regions with biological precision.
- Pairwise identities ranged from 76.0% (Dog vs Mouse) to 87.1% (Human vs Dog), confirming high overall conservation consistent with strong functional constraint.

- Two regions showed 100% conservation across all 5 species: (1) AHGKKV (F-helix, proximal His region) and (2) SELHCDKLHVDPEN (His F8 + Asp FG1 block) both directly involved in haem-iron coordination and oxygen binding. These are the most functionally critical regions of HBB.
- Variable regions (N-terminus, BC loop, EF loop) are surface-exposed and functionally tolerant, consistent with neutral evolutionary drift.
- Mouse HBB shows the lowest identity with all other species (~76–78%), reflecting the elevated substitution rate of rodents rather than greater evolutionary distance.
- All findings align with the known 3D structure of human HbA and have direct clinical applications in haemoglobinopathy research, drug development, and HBOC design.

This task has provided hands-on experience with a core bioinformatics workflow from sequence retrieval through alignment to structural interpretation reinforcing the critical connection between computational biology and clinical biomedical science.